

MTH-262 Statistics and Probability Theory

Lecture 14



Outline of this Lecture

- Mathematical Expectation
 - of discrete and
 - continuous random variables.
- Mathematical Expectation of functions
 - of continuous and
 - discrete random variables.
- Variance
 - of the discrete and
 - continuous random variables.

Mean of a Random Variable

- Previously, we discussed the sample mean, which is the arithmetic mean of the data. Now consider the following.
- If two coins are tossed 16 times and X is the number of heads that occur per toss, then the values of X are 0, 1, and 2.
- Suppose that the experiment yields no heads, one head, and two heads a total of 4, 7, and 5 times, respectively.
- The average number of heads per toss of the two coins is then

$$\frac{(0)(4) + (1)(7) + (2)(5)}{16} = 1.06.$$

This is an average value of the data and yet it is not a possible outcome of $\{0, 1, 2\}$.

Mean of a Random Variable

- Hence, an average is not necessarily a possible outcome for the experiment.
- For instance, a salesman's average monthly income is not likely to be equal to any of his monthly paychecks.
- Let us now restructure our computation for the average number of heads so as to have the following equivalent form:

$$(0) \left(\frac{4}{16} \right) + (1) \left(\frac{7}{16} \right) + (2) \left(\frac{5}{16} \right) = 1.06.$$

- The numbers 4/16, 7/16, and 5/16 are the fractions of the total tosses resulting in 0, 1, and 2 heads, respectively.
- These fractions are also the relative frequencies for the different values of X in our experiment.

Mathematical Expectations

Let X be a random variable with probability distribution $f(x)$. The mean, or expected value, of X is

$$\mu = E(X) = \sum_x x f(x)$$

if X is discrete, and

$$\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

if X is continuous.

Mathematical Expectation

- The way to calculate the expected value, or mean, shown here is different from the way to calculate the sample mean described in Chapter 1, where the sample mean is obtained by using data.
- In mathematical expectation, the expected value is calculated by using the probability distribution.
- However, the mean is usually understood as a “center” value of the underlying distribution if we use the expected value, as in Definition

Mathematical Expectation of discrete random variable (Example)

Question: A lot containing 7 components is sampled by a quality inspector; the lot contains 4 good components and 3 defective components. A sample of 3 is taken by the inspector. Find the expected value of the number of good components in this sample.

Solution:

Given information: Total Components = 7

Good Components = 4

Defective Components = 3

Number of units selected as a sample = 3

Variable of Interest = Number of good components = X

Mathematical Expectation of discrete random variable (Example)

Solution: Let X represent the number of good components in the sample. The probability distribution of X is

$$f(x) = \frac{\binom{4}{x} \binom{3}{3-x}}{\binom{7}{3}}, \quad x = 0, 1, 2, 3.$$

Simple calculations yield $f(0) = 1/35$, $f(1) = 12/35$, $f(2) = 18/35$, and $f(3) = 4/35$. Therefore,

$$\mu = E(X) = (0) \left(\frac{1}{35} \right) + (1) \left(\frac{12}{35} \right) + (2) \left(\frac{18}{35} \right) + (3) \left(\frac{4}{35} \right) = \frac{12}{7} = 1.7.$$

Thus, if a sample of size 3 is selected at random over and over again from a lot of 4 good components and 3 defective components, it will contain, on average, 1.7 good components.

Mathematical Expectation of discrete random variable (Example)

Question: A salesperson for a medical device company has two appointments on a given day. At the first appointment, he believes that he has a 70% chance to make the deal, from which he can earn \$1000 commission if successful. On the other hand, he thinks he only has a 40% chance to make the deal at the second appointment, from which, if successful, he can make \$1500. What is his expected commission based on his own probability belief? Assume that the appointment results are independent of each other.

Solution:

First, we know that the salesperson, for the two appointments, can have 4 possible commission totals: \$0, \$1000, \$1500, and \$2500. We then need to calculate their associated probabilities. By independence, we obtain

Mathematical Expectation of discrete random variable

(Example)

Solution:

First, we know that the salesperson, for the two appointments, can have 4 possible commission totals: \$0, \$1000, \$1500, and \$2500. We then need to calculate their associated probabilities. By independence, we obtain

$$f(\$0) = (1 - 0.7)(1 - 0.4) = 0.18, f(\$2500) = (0.7)(0.4) = 0.28,$$

$$f(\$1000) = (0.7)(1 - 0.4) = 0.42, \text{ and } f(\$1500) = (1 - 0.7)(0.4) = 0.12.$$

Therefore, the expected commission for the salesperson is:

$$\begin{aligned} E(X) &= (\$0)(0.18) + (\$1000)(0.42) + (\$1500)(0.12) + (\$2500)(0.28) \\ &= \$1300. \end{aligned}$$

Mathematical Expectation of continuous random variable (Example)

Question: Let X be the random variable that denotes the life in hours of a certain electronic device. The probability density function is

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected (mean) life of this type of device.

Mathematical Expectation of continuous random variable (Example)

Solution: Given information, life in hours of a certain electronic device = X

$$f(x) = \begin{cases} \frac{20,000}{x^3}, & x > 100, \\ 0, & \text{elsewhere.} \end{cases}$$

We know that $\mu = E(X) = \int_{-\infty}^{\infty} x f(x) dx$

We have
$$\mu = E(X) = \int_{100}^{\infty} x \frac{20,000}{x^3} dx = \int_{100}^{\infty} \frac{20,000}{x^2} dx = 200.$$

Therefore, we can expect this type of device to last, on average, 200 hours.

Expected Value of the Random Variable $g(X)$

Let X be a random variable with probability distribution $f(x)$. The expected value of the random variable $g(X)$ is

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x)$$

if X is discrete, and

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) \, dx$$

if X is continuous.

Expected Value of the Random Variable $g(X)$

(Example)

Question: Suppose that the number of cars X that pass through a car wash between 4:00 P.M. and 5:00 P.M. on any sunny Friday has the following probability distribution:

x	4	5	6	7	8	9
$P(X = x)$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Let $g(X) = 2X - 1$ represent the amount of money, in dollars, paid to the attendant by the manager. Find the attendant's expected earnings for this particular time period.

Expected Value of the Random Variable $g(X)$ (Example)

Solution: By definition:

$$\mu_{g(X)} = E[g(X)] = \sum_x g(x)f(x)$$

So, the expected earnings are:

$$\begin{aligned} E[g(X)] &= E(2X - 1) = \sum_{x=4}^9 (2x - 1)f(x) \\ &= (7) \left(\frac{1}{12} \right) + (9) \left(\frac{1}{12} \right) + (11) \left(\frac{1}{4} \right) + (13) \left(\frac{1}{4} \right) \\ &\quad + (15) \left(\frac{1}{6} \right) + (17) \left(\frac{1}{6} \right) = \$12.67. \end{aligned}$$

Expected Value of the Random Variable $g(X)$ (Example)

Question: Let X be a random variable with density function

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = 4X + 3$.

Expected Value of the Random Variable $g(X)$ (Example)

Solution: By definition,

$$\mu_{g(X)} = E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) \, dx$$

We have

$$\begin{aligned} E(4X + 3) &= \int_{-1}^2 \frac{(4x + 3)x^2}{3} \, dx \\ &= \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) \, dx = 8. \end{aligned}$$

So, the expected (mean) value for the given density function is 8.

Expected Value, of the Random Variable $g(X, Y)$

Let X and Y be random variables with joint probability distribution $f(x,y)$. The mean, or expected value, of the random variable $g(X, Y)$ is

$$\mu_{g(X,Y)} = E[g(X,Y)] = \sum_x \sum_y g(x,y) f(x,y)$$

If X and Y are discrete, and

$$\mu_{g(X,Y)} = E[g(X,Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x,y) f(x,y) dx dy$$

if X and Y are continuous.

Expected Value of the Random Variable $g(X,Y)$ (Example)

Question: Let X and Y be the random variables with joint probability distribution indicated in Table below:

$f(x, y)$		x			Row Totals
		0	1	2	
y	0	$\frac{3}{28}$	$\frac{9}{28}$	$\frac{3}{28}$	$\frac{15}{28}$
	1	$\frac{3}{14}$	$\frac{3}{14}$	0	$\frac{3}{7}$
	2	$\frac{1}{28}$	0	0	$\frac{1}{28}$
Column Totals		$\frac{5}{14}$	$\frac{15}{28}$	$\frac{3}{28}$	1

Find the expected value of $g(X, Y) = XY$.

Expected Value of the Random Variable $g(X,Y)$ (Example)

Solution: By definition, we can write

$$\begin{aligned} E(XY) &= \sum_{x=0}^2 \sum_{y=0}^2 xyf(x,y) \\ &= (0)(0)f(0,0) + (0)(1)f(0,1) \\ &\quad + (1)(0)f(1,0) + (1)(1)f(1,1) + (2)(0)f(2,0) \\ E(XY) &= \frac{3}{14}. \end{aligned}$$

The expected value of the variable $g(X,Y)=XY$ is $3/14$.

Expected Value of the Random Variable $g(X,Y)$

(Example)

- Question: It is known that the ratio of gallium to arsenide does not affect the functioning of gallium-arsenide wafers, which are the main components of microchips. Let X denote the ratio of gallium to arsenide and Y denote the functional wafers retrieved during a 1-hour period. X and Y are independent random variables with the joint density function

$$f(x,y) = \begin{cases} \frac{x(1+3y^2)}{4}, & 0 < x < 2, 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- Show that $E(XY) = E(X)E(Y)$,

Expected Value of the Random Variable $g(X,Y)$

(Example)

- By definition,

$$E(XY) = \int_0^1 \int_0^2 \frac{x^2 y (1 + 3y^2)}{4} dx dy = \frac{5}{6}, \quad E(X) = \frac{4}{3}, \quad \text{and} \quad E(Y) = \frac{5}{8}.$$

- Hence,

$$E(X)E(Y) = \left(\frac{4}{3}\right) \left(\frac{5}{8}\right) = \frac{5}{6} = E(XY).$$

Variance of the Random Variable

- Let X be a random variable with probability distribution $f(x)$ and mean μ . The variance of X is

$$\sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 f(x), \quad \text{if } X \text{ is discrete, and}$$

$$\sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx, \quad \text{if } X \text{ is continuous.}$$

- The positive square root of the variance, σ , is called the standard deviation of X .

Variance of the Random Variable (Example)

Question: The weekly demand for a drinking-water product, in thousands of liters, from a local chain of efficiency stores is a continuous random variable X having the probability density:

$$f(x) = \begin{cases} 2(x - 1), & 1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the mean and variance of X .

Variance of the Random Variable (Example)

Solution: Calculating $E(X)$ and $E(X^2)$, we have

$$\mu = E(X) = 2 \int_1^2 x(x-1) dx = \frac{5}{3}$$

$$E(X^2) = 2 \int_1^2 x^2(x-1) dx = \frac{17}{6}.$$

Therefore,

$$\sigma^2 = \frac{17}{6} - \left(\frac{5}{3}\right)^2 = \frac{1}{18}.$$

At this point, the variance or standard deviation has meaning only when we compare two or more distributions that have the same units of measurement.

Variance of the Random Variable

- Let X be a random variable with probability distribution $f(x)$. The variance of the random variable $g(X)$ is

$$\sigma_{g(X)}^2 = E\{[g(X) - \mu_{g(X)}]^2\} = \sum_x [g(x) - \mu_{g(X)}]^2 f(x)$$

if X is discrete, and

$$\sigma_{g(X)}^2 = E\{[g(X) - \mu_{g(X)}]^2\} = \int_{-\infty}^{\infty} [g(x) - \mu_{g(X)}]^2 f(x) dx$$

if X is continuous.

Variance of the Random Variable (Example)

- Question: Calculate the variance of $g(X) = 2X + 3$, where X is a random variable with probability distribution

x	0	1	2	3
$f(x)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{2}$	$\frac{1}{8}$

- To solve above question, we need to find a new variable $g(X)$. And the probability of each value of $g(X)$ will be same as the probability of the value of X .

Variance of the Random Variable (Example)

- First, we find the mean of the random variable $2X + 3$.

$$\mu_{2X+3} = E(2X + 3) = \sum_{x=0}^3 (2x + 3)f(x) = 6.$$

- Now, the variance of $g(X)$ is:

$$\begin{aligned}\sigma_{2X+3}^2 &= E\{[(2X + 3) - \mu_{2X+3}]^2\} = E[(2X + 3 - 6)^2] \\ &= E(4X^2 - 12X + 9) = \sum_{x=0}^3 (4x^2 - 12x + 9)f(x) = 4.\end{aligned}$$

Variance of the Random Variable (Example)

- Question: Let X be random variable having the density function:

$$f(x) = \begin{cases} \frac{x^2}{3}, & -1 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the variance of the random variable $g(X) = 4X + 3$.

- The expected (mean) value for the given density function is 8.

Variance of the Random Variable (Example)

- Solution: The expectation and variance of $g(X)=4X+3$

$$E(4X + 3) = \int_{-1}^2 \frac{(4x + 3)x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (4x^3 + 3x^2) dx = 8.$$

$$\begin{aligned} \sigma_{4X+3}^2 &= E\{[(4X + 3) - 8]^2\} = E[(4X - 5)^2] \\ &= \int_{-1}^2 (4x - 5)^2 \frac{x^2}{3} dx = \frac{1}{3} \int_{-1}^2 (16x^4 - 40x^3 + 25x^2) dx = \frac{51}{5}. \end{aligned}$$

Covariance

- Let X and Y be random variables with joint probability distribution $f(x,y)$. The covariance of X and Y is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \sum_x \sum_y (x - \mu_X)(y - \mu_Y) f(x, y)$$

if X and Y are discrete, and

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f(x, y) dx dy$$

if X and Y are continuous.

- The sign of the covariance indicates whether the relationship between two dependent random variables is positive or negative.

Correlation Coefficient

- Let X and Y be random variables with covariance σ_{XY} and standard deviations σ_X and σ_Y , respectively. The correlation coefficient of X and Y is

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

- It should be clear to the reader that ρ_{XY} is free of the units of X and Y . The correlation coefficient satisfies the inequality $-1 \leq \rho_{XY} \leq 1$. It assumes a value of zero when $\sigma_{XY} = 0$.

Chebyshev's Theorem

- The probability that any random variable X will assume a value within k standard deviations of the mean is at least $1 - 1/k^2$. That is,

$$P(\mu - k\sigma < X < \mu + k\sigma) \geq 1 - \frac{1}{k^2}.$$

- For $k = 2$, the theorem states that the random variable X has a probability of at least $1 - 1/2^2 = 3/4$ of falling within two standard deviations of the mean. That is, three-fourths or more of the observations of any distribution lie in the interval $\mu \pm 2\sigma$. Similarly, the theorem says that at least eight-ninths of the observations of any distribution fall in the interval $\mu \pm 3\sigma$.

Chebyshev's Theorem (Example)

- A random variable X has a mean $\mu=8$, a variance $\sigma^2=9$, and an unknown probability distribution. Find
(a) $P(-4 < X < 20)$, (b) $P(|X - 8| \geq 6)$.
- Solution:
- (a) $P(-4 < X < 20) = P[8 - (4)(3) < X < 8 + (4)(3)] \geq 15/16$.
- (b) $P(|X - 8| \geq 6) = 1 - P(|X - 8| < 6) = 1 - P(-6 < X - 8 < 6)$
 $= 1 - P[8 - (2)(3) < X < 8 + (2)(3)] \leq 1/4$

Activity 1

- Question: The total time, measured in units of 100 hours, that a teenager runs her hair dryer over a period of one year is a continuous random variable X that has the density function

$$f(x) = \begin{cases} x, & 0 < x < 1, \\ 2 - x, & 1 \leq x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- Find the mean of the random variable $Y = 60X^2 + 39X$, where Y is equal to the number of kilowatt hours expended annually.

Activity 2

- Question: The distribution of times before a major repair of a washing machine was given as

$$f(y) = \begin{cases} \frac{1}{4}e^{-y/4}, & y \geq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

What is the population mean of the times to repair?

- Question: A continuous random variable X has the density function

$$f(x) = \begin{cases} e^{-x}, & x > 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Find the expected value of $g(X) = e^{2X/3}$.

Properties of Expectation

- Expectation

(i) $E(aX + b) = aE(X) + b,$ $\therefore E(b) = b.$

(ii) $E[g(X) \pm h(X)] = E[g(X)] \pm E[h(X)].$

(iii) $E[g(X, Y) \pm h(X, Y)] = E[g(X, Y)] \pm E[h(X, Y)].$

(iv) $E[X \pm Y] = E[X] \pm E[Y].$

(v) $E(XY) = E(X)E(Y).$ $\therefore X \& Y$ are Independent

(vi) $E(Z) = E(X/Y) \neq E(X)/E(Y),$

Variance of the Linear Combination of Variables

- If X and Y are random variables with joint probability distribution $f(x,y)$ and a , b , and c are constants, then

$$(i) \quad \sigma^2_{aX+bY+c} = a^2\sigma^2_X + b^2\sigma^2_Y + 2ab\sigma_{XY} .$$

$$(ii) \quad \sigma^2_{aX+c} = a^2\sigma^2_X = a^2\sigma^2. \quad \therefore b=0$$

$$(iii) \quad \sigma^2_{X+c} = \sigma^2_X = \sigma^2. \quad \therefore a=1, b=0, \text{Var}(c)=0$$

$$(iv) \quad \sigma^2_{aX+bY} = a^2\sigma^2_X + b^2\sigma^2_Y .$$